



CAPABILITY
PORTFOLIO
2026

AI · DATA · EARTH OBSERVATION ·

Seeing Earth, Smarter

AI, data and Earth Observation solutions that turn complex information into actionable intelligence for governments, businesses and organizations.

PARIS, FRANCE · BEIRUT, LEBANON

[APP.RASID.AI](#) · [RASID.AI](#)

Deep-Tech AI & Geospatial Consultancy

RASID is an innovation-led deep-tech company turning your data into decision-ready intelligence. We combine deep geospatial expertise, AI, data and software engineering to build production systems for governments, enterprises and international organizations.

01

11 years

11+ years of combined team experience across remote sensing, deep learning, data engineering and problem-solving.

02

Bespoke Consultancy

We work directly with organizations to design and deliver tailored geospatial, data engineering and AI solutions.

03

GoPilot

Our flagship geospatial AI platform, turning natural-language questions into executable Earth Observation workflows.

10,000+

DATASETS AVAILABLE THROUGH GOPILOT

04

R&D Experience

Our R&D spans AI, remote sensing, data engineering and GeoAI, with 60+ scientific publications and a focus on turning validated methods into operational solutions.

05

AWS GenAI Challenge Winner

GoPilot was selected as a winner of the AWS Geospatial GenAI Challenge 2026, recognizing our work at the intersection of AI and Earth Observation.

Services

RASID combines AI, data engineering and Earth observation to build production-ready solutions for climate, environmental, agricultural and geospatial applications.

01

AI Engineering & Consultancy

From AI strategy and model development to production deployment, we design and integrate AI systems across agentic AI, computer vision, RAG, OCR and MLOps. We build and integrate AI systems that turn complex data and models into practical, production-ready solutions..

02

Agent Design & Development

We design and build production-ready AI agents that connect models, tools, data and knowledge to automate complex workflows. From agent architecture and tool orchestration to RAG, integrations and deployment, we develop tailored agents around specific business and operational needs.

03

Data Engineering

Design and development of scalable data pipelines, from ingestion and transformation to processing, storage and delivery. We integrate structured and unstructured data to support analytics, AI applications and operational workflows.

04

Climate & Environmental Intelligence

Climate and environmental monitoring using Earth observation and multi-source data, including methane detection, climate-risk assessment, wildfire and flood monitoring, landscape change, urban heat, land-use change and settlement dynamics.

05

AgriTech

Crop monitoring, crop mapping, wheat and banana analytics, grain-bin classification and yield forecasting.

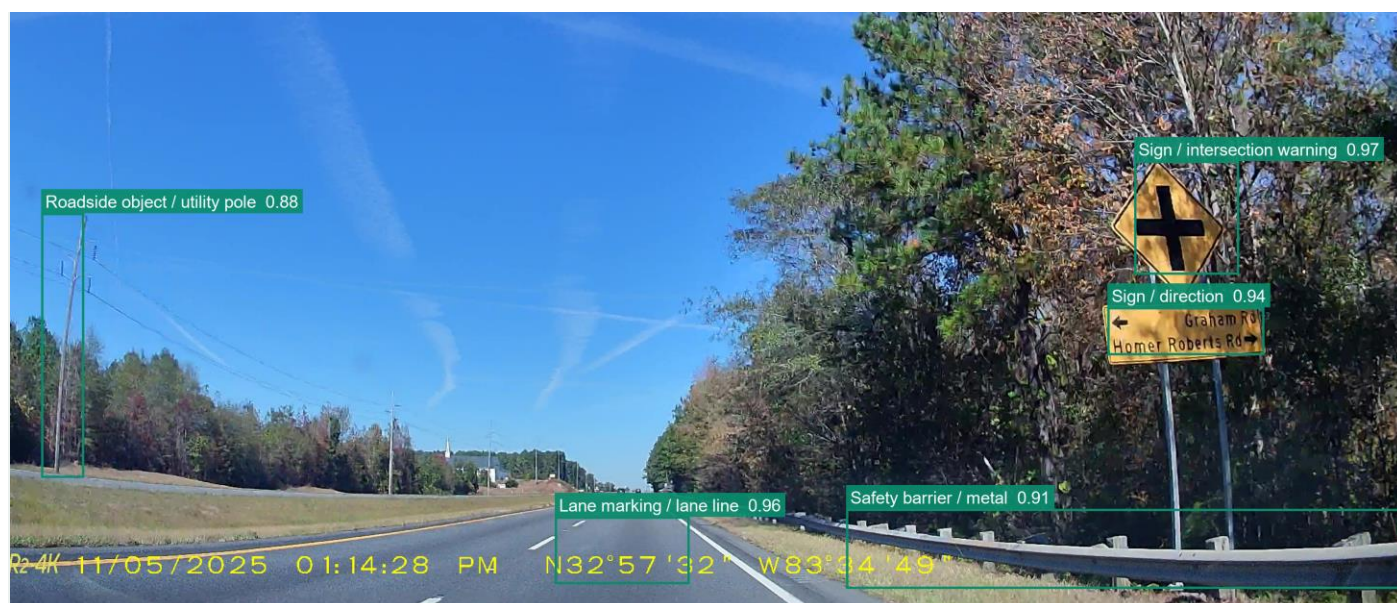
06

Analysis-Ready Satellite Data

Raw satellite imagery processing, quality assessment, anomaly detection, geometric and radiometric correction, orthorectification and analysis-ready data preparation.

International Consulting Firm · Road Safety · 2026

iRAP Road Coding Automation



OVERVIEW

A data engineering and computer vision pipeline that extracts road attributes from georeferenced survey imagery to drive iRAP star-rating coding.

SOLUTION (METHODOLOGY)

Imagery ingestion and georeferencing, then detection and segmentation models for lane markings, roadside hazards, signage and footpaths.

PROBLEM

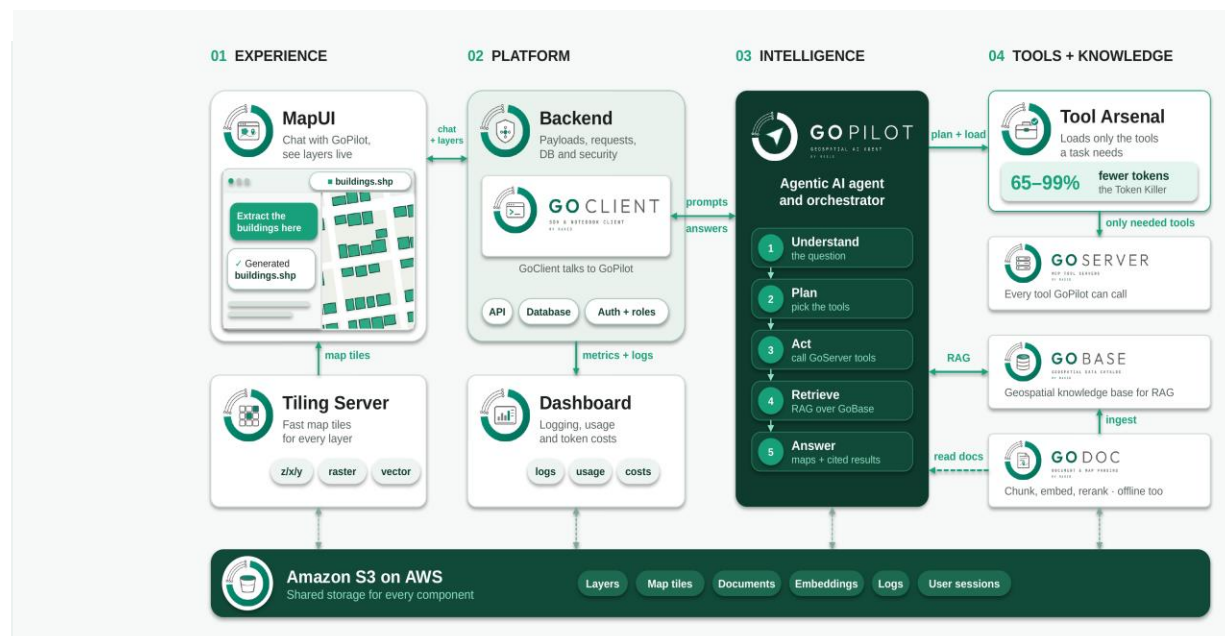
iRAP coding is manual and slow. Analysts review survey imagery frame by frame to code dozens of attributes per segment, which caps network coverage.

RESULTS

55x speed up in operations, 2,400 km of coded network, 1.2 million survey frames processed

PRODUCT PLATFORM · AGENTIC GEOAI SUITE

GoSuite: One Agent, Every Geospatial Tool



OVERVIEW

GoSuite is RASID's end to end platform for agentic GeoAI. GoPilot, the agent and orchestrator, sits at the center and works with GoServer tools, GoBase knowledge and GoDoc document intelligence, reaching users through MapUI and developers through GoClient.

SOLUTION (METHODOLOGY)

MapUI chats with GoPilot through a secure Backend (GoClient, database, auth) while a Tiling Server streams layers to the map. GoPilot plans each task and the Tool Arsenal loads only the GoServer tools it needs. GoDoc chunks, embeds and reranks documents into GoBase for RAG, and every component shares one S3 store.

PROBLEM

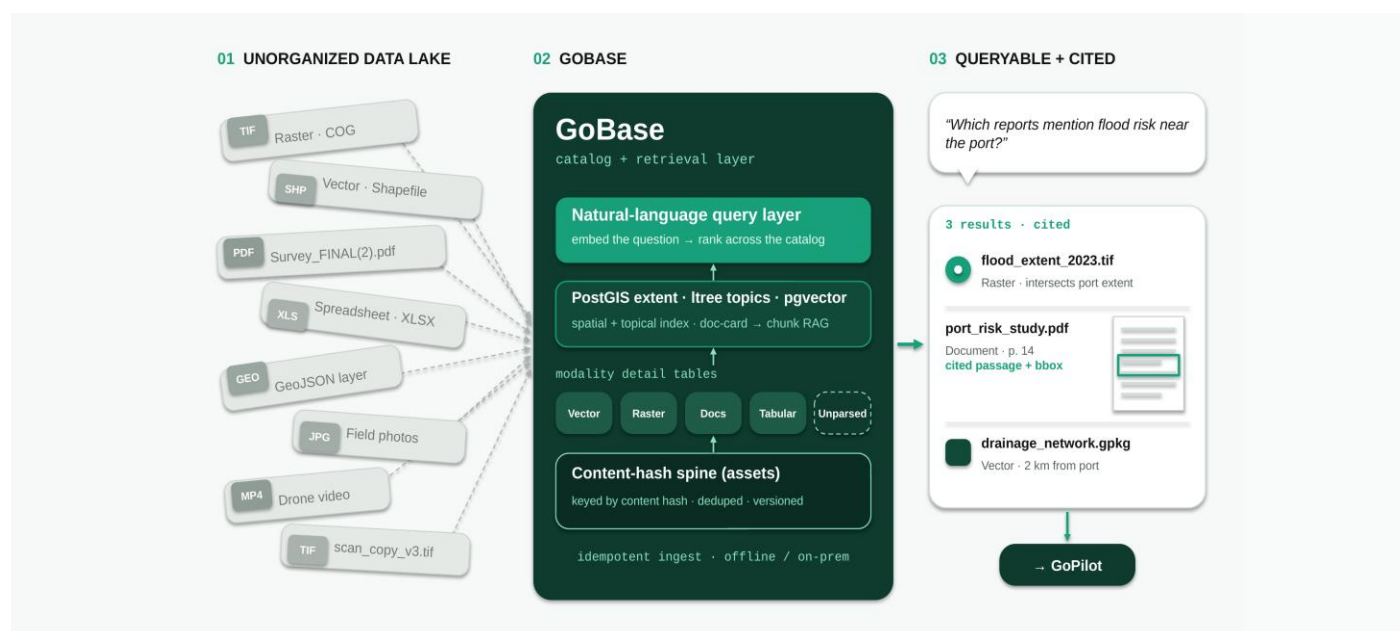
A geospatial answer usually takes several tools, scattered data and an expert to stitch them together. Handing that to an LLM agent naively is expensive: loading every tool definition into every request burns tokens and slows the agent down.

RESULTS

A simple request like "extract the buildings here" becomes a finished layer on the map, backed by cited sources. The Tool Arsenal cuts token costs by 65 to 99%, and dashboards and logs track usage and cost across the whole suite.

DATA INFRASTRUCTURE · GEOSPATIAL CATALOG & RETRIEVAL

GoBase: A Queryable Catalog for Data Lakes



OVERVIEW

A catalog and retrieval layer that indexes an entire geospatial data lake (raster, vector, documents, spreadsheets, images and video) into one queryable spine with a natural-language layer on top.

SOLUTION (METHODOLOGY)

A content-addressed spine (deduped, version-tracked) with a detail table per modality, including an "unparsed" catch-all. PostGIS extents and an Itree topic tree index space and subject; a two-level pgvector RAG store finds the document, then the cited page and bbox.

PROBLEM

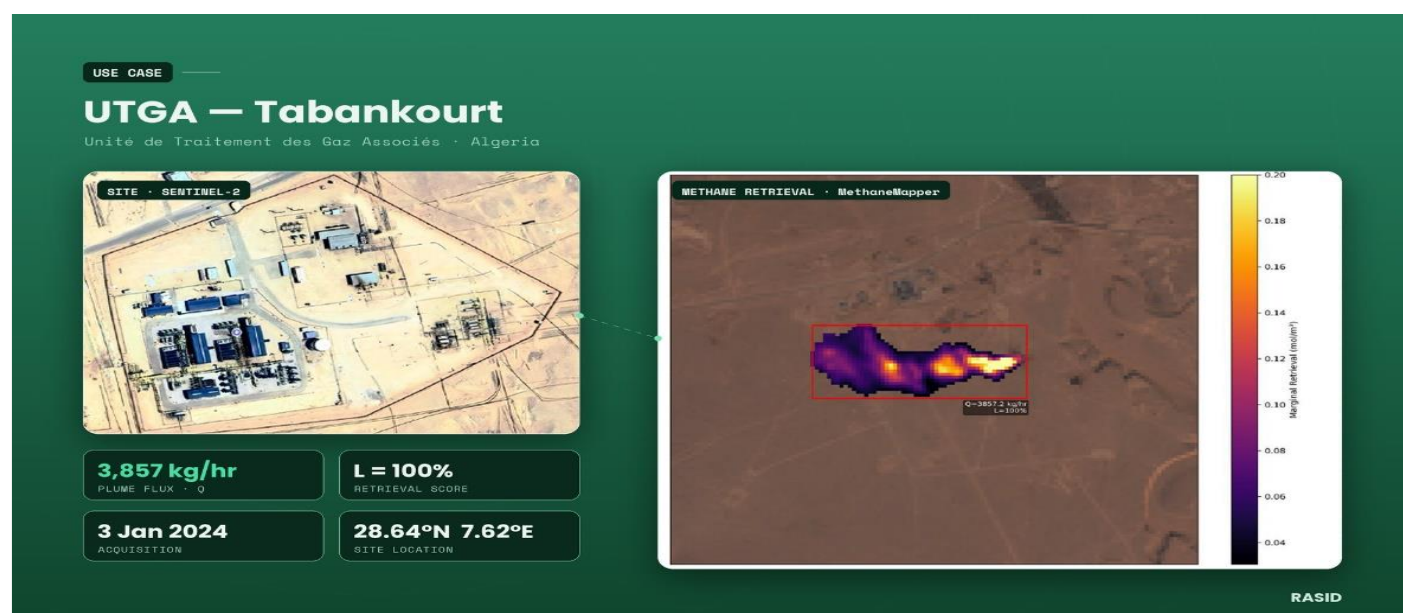
Large geospatial lakes are scattered across formats and folders, identified by filename rather than content, with no single index across modalities and no way to ask a plain-language question, so most of the data stays invisible.

RESULTS

A heterogeneous lake becomes one searchable catalog with cited answers. Ingestion is idempotent and runs offline or on-prem, and GoPilot can answer natural-language questions over data that was previously unreachable.

EBRD CLIMATE CHANGE INNOVATION · Methane Monitoring

Oil&Gas Methane Plume Monitoring



OVERVIEW

AI-based methane plume detection and quantification for oil & gas operations using satellite imagery.

SOLUTION (METHODOLOGY)

Physics-informed AI detects potential methane plumes and supports rapid site-level investigation.

PROBLEM

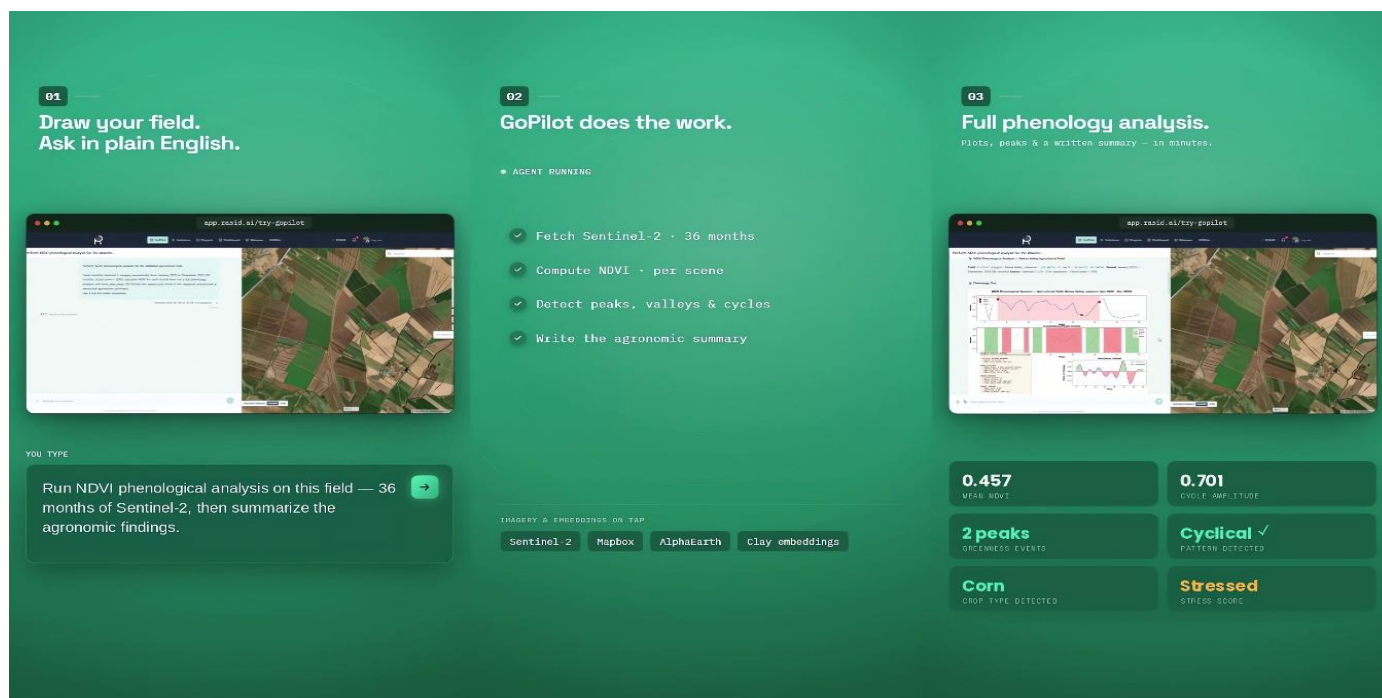
Methane leaks from oil & gas infrastructure are difficult to detect and monitor at scale.

RESULTS

80% of confirmed UNEP plume events recovered, with results delivered in minutes.

AGRITECH · CROP MONITORING · BANANA TR4

Banana Crop Monitoring



OVERVIEW

Satellite-based crop monitoring for banana plantations, with a focus on early detection of Fusarium Wilt (TR4).

SOLUTION (METHODOLOGY)

Sentinel-2 imagery is analysed to detect crop stress and potential disease signatures before visible outbreaks.

PROBLEM

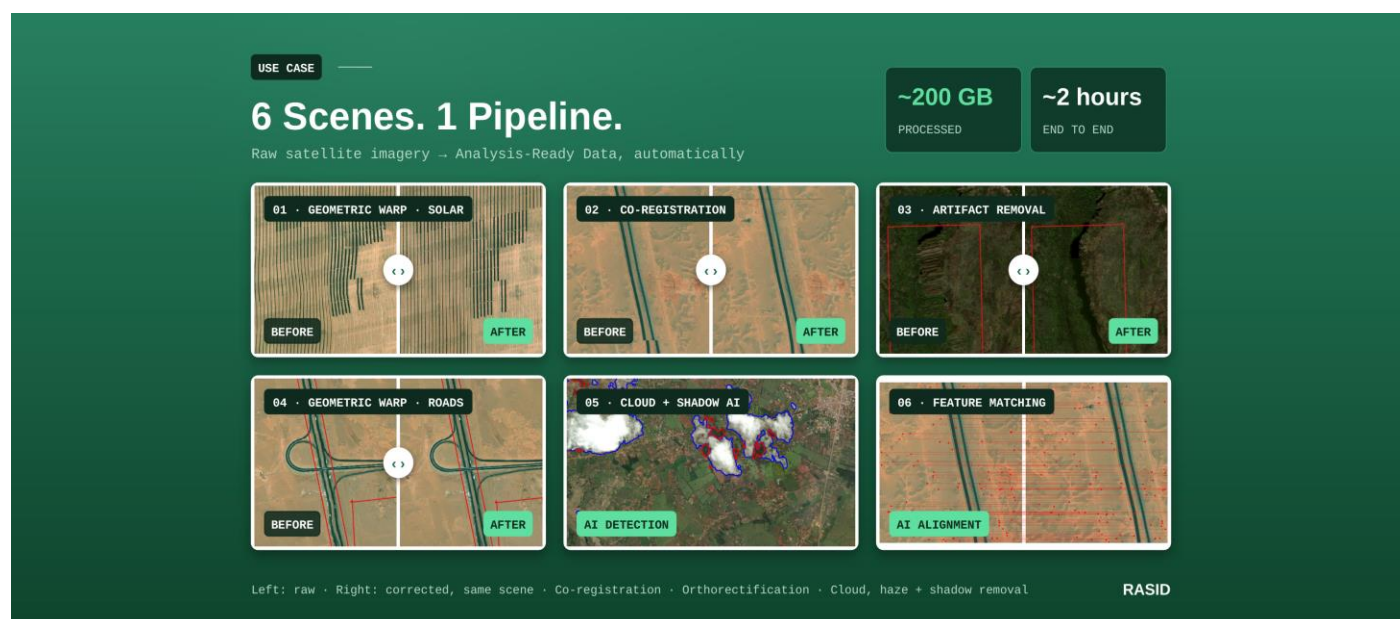
Fusarium Wilt (TR4) can devastate banana crops before visible symptoms appear, creating a need for early warning.

RESULTS

Operational crop-monitoring workflow combining phenology analysis with potential TR4 stress detection.

PRIVATE SATELLITE-IMAGERY PROVIDER · 2026

Analysis-Ready Satellite Data



OVERVIEW

AI-based pipeline that corrects geometric and radiometric anomalies in high-resolution satellite imagery and converts raw data into Analysis-Ready Data (ARD).

SOLUTION (METHODOLOGY)

Deep-learning co-registration, orthorectification, radiometric enhancement and automated cloud, haze and shadow detection and removal.

PROBLEM

Geometric distortion, radiometric inconsistencies and atmospheric artefacts can make satellite imagery difficult to use and require costly manual correction.

RESULTS

Client-ready ARD pipeline processing ~200 GB in ~2 hours, turning raw imagery into sellable products.

Projects

R&D and consulting projects delivered by a team with experience across AI and geospatial technologies.

REF	PROJECT	BENEFICIARY
P09	iRAP Road Coding Automation	International Consulting Firm
P08	Synthetic Methane Plume for Satellite-Based MRV	Brazil-based Client
P07	Oil & Gas Methane Plume Monitoring	EBRD Climate Change Innovation Programme
P06	Banana Crop Monitoring	Private Agricultural Client
P05	AWS GenAI Challenge 2026	Amazon AWS
P04	AI for Food Security in the Arab World	AOAD
P03	Grain Bins Classification across the USA	US-based AgriTech Client
P02	Handwritten OCR for the Public Sector	UNDP - RBAS
P01	RAG Knowledge Platform	Private Consultancy Client



Team

A senior team combining research, AI engineering, geospatial expertise and commercial experience directly involved in every engagement.



01

Dr. Ali J. Ghandour

AI R&D AI Consultant



02

Reda Haidar

Commercial & Strategy Director



03

Hasan Nasrallah

AI Solutions Architect



04

Hasan Wehbi

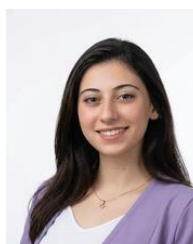
AI System Engineer



05

Mohamad El Moussawi

Platform Engineer



06

Amira El Halabi

BD & Pre-Sales



Let's Talk.

RASID delivers bespoke AI consulting, data engineering and geospatial solutions, combining technical expertise with custom development to address complex real-world challenges.

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